

BILKENT UNIVERSITY

Engineering Mathematics I Midterm 1					
Code : <i>Math 241</i>	Last Name:				
Acad. Year: <i>2017-2018</i>	Name :		Student No.:		
Semester : <i>Fall</i>	Section: :		Signature:		
Date : <i>13.11.2017</i>	4 QUESTIONS ON 4 PAGES TOTAL 100 POINTS				
Time : <i>17:40</i>					
Duration : <i>120 minutes</i>					
1	2	3	4		

No partial credit will be given for unsupported answers!

1. (30 pts) Consider the system

$$x_1 + 2x_2 + x_3 + x_4 = \alpha$$

$$2x_1 + 3x_2 + x_3 - x_4 = -1$$

$$3x_1 + 4x_2 + x_3 + x_4 = 0$$

$$x_1 + x_2 = 1$$

- Express this linear system of equations in terms of $Ax = c$ and identify entries of A , x and c .
- Apply elementary row operations to reduce the system to its reduced row echelon form.
- For which value of α the system is consistent? For this value, determine the solutions of $Ax = c$.
- Find a basis for the row space of A .
- Find a basis for the column space of A .
- Find a basis for the null space of A .

(a)
$$\underbrace{\begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 3 & 1 & -1 \\ 3 & 4 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} \alpha \\ -1 \\ 0 \\ 1 \end{bmatrix}}_c$$

(b)
$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & \alpha \\ 2 & 3 & 1 & -1 & -1 \\ 3 & 4 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\text{ERO's}} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & -1/2 - 9\alpha/4 \\ 0 & 1 & 1 & 0 & 1/2 + 7\alpha/4 \\ 0 & 0 & 0 & 1 & -1/2 - \alpha/4 \\ 0 & 0 & 0 & 0 & 1 + \alpha/2 \end{array} \right]$$

↑ ↑ ↑
pivot columns

(c) $1 + \frac{\alpha}{2} = 0 \Rightarrow \alpha = -2$

(d) $\text{Row}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

(e) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \right\}$

Solution Set $\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} : \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \cdot \alpha + \begin{bmatrix} 4 \\ -3 \\ 0 \\ 0 \end{bmatrix}, \alpha \in \mathbb{R} \right\}$

(f) $\text{Null}(A) = \left\{ \underline{x} : \underline{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \alpha, \alpha \in \mathbb{R} \right\}, \text{ basis for } \text{Null}(A) = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$

2. (25 pts) Find eigenvalues and associated eigenspaces of the following matrix:

$$A = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 4 & 2 \\ 1 & -3 & 1 \end{bmatrix}.$$

Can A be diagonalizable?

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 3 & 2 \\ 1 & 4-\lambda & 2 \\ 1 & -3 & 1-\lambda \end{vmatrix} = (2-\lambda)[(4-\lambda)(1-\lambda)+6] - 3((1-\lambda)-2) \\ &\quad + 2(-3-4+\lambda) \\ &= (2-\lambda)(10-5\lambda+\lambda^2) + 3 + 3\lambda + 2\lambda - 14 \\ &= -\lambda^3 + 7\lambda^2 - 20\lambda + 20 + 5\lambda - 11 \\ &= -\lambda^3 + 7\lambda^2 - 15\lambda + 9 \end{aligned}$$

$$p(\lambda) = -\lambda^3 + 7\lambda^2 - 15\lambda + 9$$

$$p(\lambda) = 0 \Rightarrow \lambda^3 - 7\lambda^2 + 15\lambda - 9 = 0 \Rightarrow (\lambda-1)(\lambda^2-6\lambda+9) = (\lambda-1)(\lambda-3)^2$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = \lambda_3 = 3.$$

$$\begin{aligned} \text{(i) Eigenspace of } \lambda_1 &= \text{Null}(A - \lambda_1 I) = \text{Null}\left(\begin{bmatrix} 1 & 3 & 2 \\ 1 & 3 & 2 \\ 1 & -3 & 0 \end{bmatrix}\right) = \text{Null}\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1/3 \end{bmatrix}\right) \\ &= \text{Span}\left\{\begin{bmatrix} -1 \\ -1/3 \\ 1 \end{bmatrix}\right\} \end{aligned}$$

$$\begin{aligned} \text{(ii) Eigenspace of } \lambda_2 &= \text{Null}(A - \lambda_2 I) = \text{Null}\left(\begin{bmatrix} -1 & 3 & 2 \\ 1 & 1 & 2 \\ 1 & -3 & -2 \end{bmatrix}\right) = \text{Null}\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}\right) \\ &= \text{Span}\left\{\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\} \end{aligned}$$

Since Eigenspace of λ_2 is not equal to multiplicity of λ_2 , A is not diagonalizable.

3. (25 pts) Let A be a 3×3 symmetric matrix with real entries. Given that $\left(1, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right)$,

$\left(2, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right)$ are two eigenpairs of A ,

(a) How to choose v_3 so that $(3, v_3)$ is also an eigenpair of A .

(b) Find a matrix A that accepts the above eigenpairs. Is A unique?

(a) Since A is symmetric it has orthogonal eigenspaces. Therefore v_3 should be orthogonal to the span of $\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right\}$

This requires: $\begin{bmatrix} 0 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} v_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow v_3 = \alpha \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \alpha \neq 0.$

(b) Orthonormalization of the set of eigenvectors $\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}\right\}$

will generate Q such that $Q^T Q = I$ and $A = Q \Lambda Q^T$.

Since the set of eigenvectors are already orthogonal, $Q = \begin{bmatrix} 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{3} & -1/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$

Thus: $A = Q \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} Q^T = \frac{1}{3} \begin{bmatrix} 8 & 1 & -1 \\ 1 & 5 & -2 \\ -1 & -2 & 2 \end{bmatrix}$

This expression for A is unique.

4. (20 pts) Find the determinant and the inverse of the $n \times n$ matrix A given as

$$A = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 2 & \dots & 2 \\ 1 & 2 & 3 & \dots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \dots & n \end{bmatrix}$$

$$\left[\begin{array}{c|c} \underline{A} & \underline{I} \end{array} \right] \xrightarrow{\text{EROS}} \left[\begin{array}{c|c} \underline{I} & \underline{A}^{-1} \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 1 & \dots & 1 & 1 \\ 1 & 2 & \dots & 2 & 1 \\ 1 & 2 & 3 & \dots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \dots & n \end{array} \right]$$

$$\begin{array}{l} R_n - R_{n-1} \rightarrow R_n \\ R_{n-1} - R_{n-2} \rightarrow R_{n-1} \\ \vdots \\ R_2 - R_1 \rightarrow R_2 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & \dots & 1 & & 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 1 & -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & 1 & 0 & \dots & -1 & 1 & 0 \\ 0 & \dots & 0 & 1 & 0 & \dots & \dots & -1 & 1 \end{array} \right]$$

$$\begin{array}{l} R_1 - R_2 \rightarrow R_1 \\ R_2 - R_3 \rightarrow R_2 \end{array}$$

$$R_{n-1} - R_n \rightarrow R_{n-1}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & \dots & 0 & 2 & -1 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & -1 & 2 & -1 & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & \dots & -1 & 2 & -1 & \\ & & & & & & & -1 & 1 \end{array} \right]$$

$\underline{I} \qquad \underline{A}^{-1}$

Since none of the elementary row operations used any scaling and row exchanges

$$\boxed{\det(\underline{A}) = \det(\underline{I}) = 1}$$